Reg. No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION ….

Third Semester

MODEL QUESTION PAPER

Minor Course - MG3DSCSTA202–

DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

(2024 ADMISSION)

PRACTICAL EXAMINATION

Duration: 1 Hour Maximum Marks: 30

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Instructions:
From the eight questions provided, students need to attempt **any five**.
Each question carries **6 marks**.
Use **statistical software — either R or Python** — to solve the questions.
Be sure to clearly write the **code/s used**, along with **important steps** and the **final answers** in your **answer script**.

[Learning Domain][CO No(s)]

1. Based on the following data, test whether the preference for Car models A and B is independent of the age groups. [E][5]

Car model / Age group	Model A	Model B
Age I	215	135
II	325	175
III	60	90

2. Based on the following data on length (X) and weight (Y), find rank correlation coefficient. [E][5]

X	70	48	58	55	54	50	60	52
Y	62	47	53	60	55	68	51	48

3. Based on the following data, obtain the regression equation of X on Y and the regression equation of Y on X. [E][5]

X	2	4	6	8	10
Y	5	7	9	8	11

4. A sample of size 8 from a normal population with standard deviation 3 is found to have mean 9. Examine whether the mean of the population is 7. [An][5]
5. Two animal foods A and B are given and the following results on weights were observed. Assuming that identical sets of animals were used in both the sets, can we conclude that food B is better than food A? [E][5]

   Food A: 49, 53, 51, 52, 47, 50, 52, 53

   Food B: 52, 55, 52, 53, 50, 54, 54, 53
6. A researcher believes that the average daily screen time of college students is more than 5 hours. A random sample of 100 students on daily screen time is collected and given below. The sample shows an average screen time of 5.428 hours with a standard deviation of 0.4758 hours. Test his claim at the 5% significance level. [E][5]

   5.2, 6.1, 4.9, 5.3, 5.7, 5.5, 4.8, 5.4, 5.6, 4.7,
   6.0, 5.1, 5.8, 5.9, 4.5, 5.3, 6.2, 5.0, 4.6, 5.6,
   5.5, 6.4, 5.0, 4.8, 5.1, 5.7, 6.1, 5.3, 5.9, 5.2,
   5.0, 5.6, 5.4, 5.3, 6.3, 4.9, 5.8, 5.7, 4.6, 5.2,
   6.0, 5.5, 5.1, 5.4, 6.1, 4.7, 5.6, 4.8, 5.3, 5.9,
   6.2, 5.0, 4.5, 5.7, 5.3, 6.0, 5.6, 5.8, 4.9, 5.5,
   6.3, 5.1, 5.7, 5.0, 5.8, 6.1, 5.3, 4.8, 5.6, 4.9,
   5.7, 5.9, 5.4, 5.1, 6.0, 5.3, 4.7, 5.2, 6.2, 5.0,
   5.5, 5.6, 5.3, 4.6, 6.1, 5.8, 5.2, 4.9, 5.7, 5.3,
   5.0, 6.3, 5.6, 5.4, 5.1, 4.8, 6.0, 5.5, 5.2, 5.9
7. A test of 100 youths and 200 adults showed that 42 of the youths and 50 of the adults were poor drivers. Use the data to test the claim that the youth percentage of poor drivers is larger than the adult percentage. [An][5]
8. The three samples given below have been obtained from three normal populations with equal variances. Test whether the population means are equal. [E][5]

$\mathbf{Y_1}$	$\mathbf{Y_2}$	$\mathbf{Y_3}$
8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

(5x6=30 marks)